9	Ophthalmic	Given into the eye by drops, gel, or ointment
10	Oral	Swallowed by mouth as a tablet, capsule, lozenge, or liquid
11	Otic	Given by drops into the ear
12	Rectal	Inserted into the rectum
13	Subcutaneous	Injected just under the skin
14	Sublingual	Held under the tongue
15	Topical	Applied to the skin
16	Transdermal	Given through a patch placed on the skin

9.11.1 Pros and Cons of different routes of Administration of Drugs

Sl. No.	Route	Advantage	Disadvantage
1	Oral	➢ Easy ➢ Preferred by patients ➢ Slow-release preparations may be available to extend duration of action ➢ Drugs can be formulated in such a way as to protect them from digestive enzymes, acid, etc.	➢ Unsuitable in patients who are uncooperative, strictly "nil by mouth", are vomiting profusely or have ileus ➢ Most orally administered drugs are absorbed slowly ➢ Unpredictable absorption due to degradation by stomach acid and enzymes
2	Rectal	➢ Good absorption – the haemorrhoidal veins drain directly into the inferior vena cava, avoiding hepatic first pass metabolism	➢ May not be suitable after rectal or anal surgery ➢ Some patients dislike suppositories
3	Subcutaneous or Intramuscular	➢ Good absorption, especially for drugs with a low oral bioavailability ➢ Onset is more rapid than the above routes ➢ Depending on formulation can have very long duration of action, e.g. depot antipsychotics and contraceptives	➢ Absorption may still be unpredictable if peripheries are poorly perfused ➢ Injections hurt, cause bruises and frighten children and needle phobics

XII_Nursing_Vocational_Unit 9.indd 252 1/11/2020 4:11:50 PM